Azure PTU Batch Accelerator

01 / Asynchronous job flow — official OpenAI SDK, existing Azure OpenAI PTUs

BLUE SDK / inference · PURPLE asynchronous work · TEAL data access

Client and identity

Entra ID

Token for the accelerator API

Entra token

Official OpenAI SDK

Custom URL + token scope

Files + Batches
Upload · submit · poll
Cancel · download

Client workflow stays familiar

1 Upload + submit

10 Poll / download

Customer-deployed accelerator

Azure Service Bus Standard · one namespace

Control queue

Validate / finalize notifications

Inference queue

Row IDs + Blob references

Azure Container Apps environment

One environment · three applications

4 Publish events

Validate / finalize

5 Work items

6 Dispatch

API application

FastAPI on Container Apps

Files + Batch API
Auth and ownership
Stream uploads / downloads

Public ingress · scales 0–2

Control Functions

Native Functions on Container Apps

Outbox relay · whole-file validation
Finalization · recovery / retention

Internal ingress · one warm replica

Inference Functions

Native Functions on Container Apps

Claim one row
Call PTU · commit outcome

Internal · scales 0–1 · 8 calls

2 Store JSONL + job / outbox

9 Read / finalize output files

3 Cosmos change feed

8 Commit outcomes

Shared application data

All three apps use managed identity · private access in the Terraform topology

Azure Blob Storage

Input JSONL · result fragments · output/error JSONL

Azure Cosmos DB serverless

Files · jobs · requests · attempts · outbox

7 PTU inference

Existing customer resource

Azure OpenAI PTU

Customer capacity already purchased

Deployment name from JSONL
Chat Completions / Responses
Ordinary inference API

Customer-managed PTU capacity

SDK contract

JSONL input · unique custom_id
Chat Completions + stateless Responses

Customer controls submission time

No off-peak scheduler or utilization controller
Existing PTU capacity must be configured and available

Durable outcomes

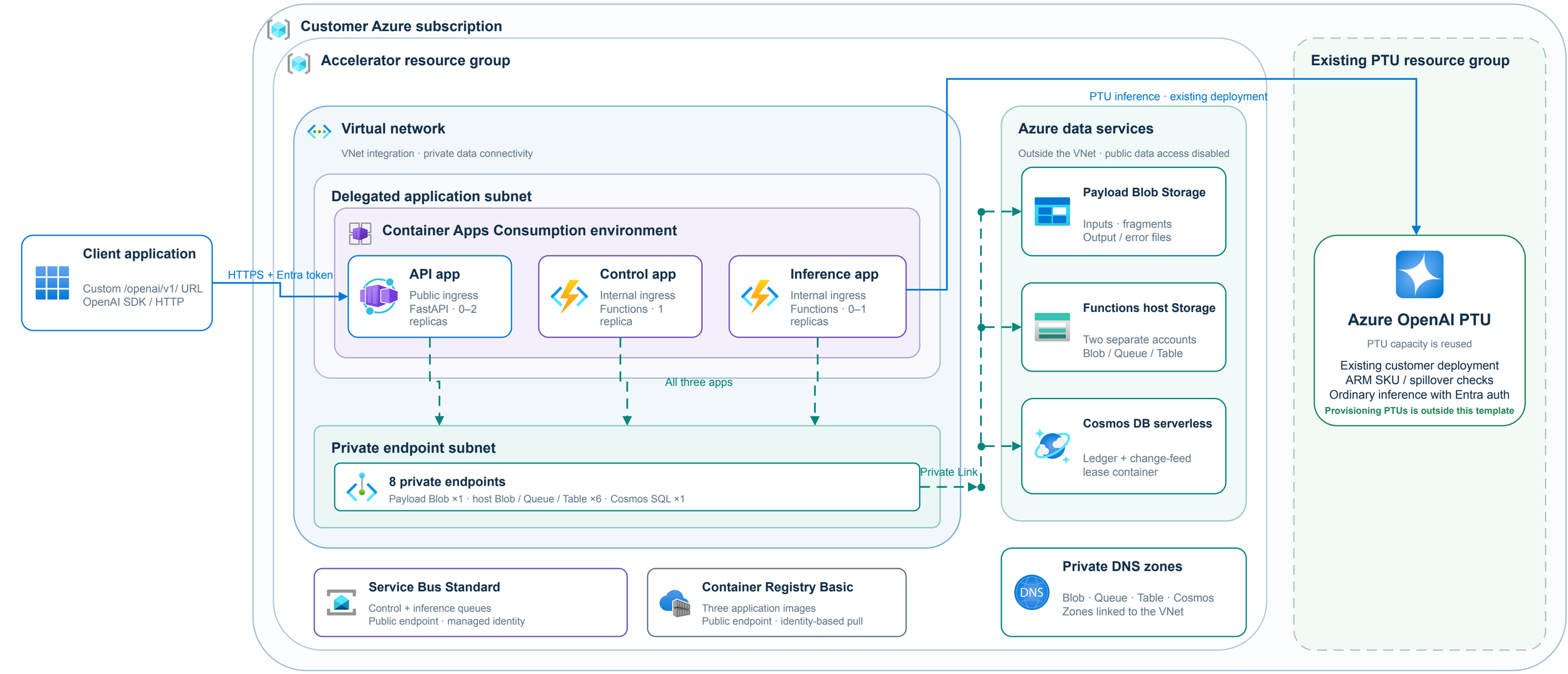
One published outcome per request
Unknown remote outcomes are not replayed by default

Reference architecture · Azure-native Files and Batch compatibility · Azure icons supplied by the user

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02 / Deployment and security — native Terraform private-networking topology

BLUE SDK / inference · PURPLE asynchronous work · TEAL data access



Microsoft Entra ID

API token audience + delegated scope / app role
Files and jobs remain isolated by caller

One user-assigned managed identity

Attached to all three apps · scoped service RBAC
No storage keys or registry admin credentials

Monitoring and operations

Application Insights + Log Analytics
Control recovery timer · guarded worker upgrades

Reference architecture · Azure-native Files and Batch compatibility · Azure icons supplied by the user